

TABLE I: NDCG performance of L2RAC compared to random ranking for neural architecture search. The cell value $\alpha \pm \beta\%$ indicates that L2RAC achieves α on NDCG, is $+\beta\%$ better or $-\beta\%$ worse than random ranking. Purple cells denote performance improvement, uncolored cells indicate no performance improved compared to random ranking. $\dagger$ and $*$ denote $p < 0.001$ and $0.001 \leq p < 0.05$, respectively.

Dataset	NDCG@1	NDCG@3	NDCG
c10mop10_0	0.2857 ^{-48.7%*}	0.7088 ^{+17.5%}	0.8408 ^{+1.3%}
c10mop11_0	0.4286 ^{-14.3%}	0.4534 ^{-24.3%*}	0.7776 ^{-6.6%*}
c10mop12_0	0.1429 ^{-77.8%*}	0.6335 ^{-10.8%}	0.7913 ^{-8.8%*}
c10mop13_0	0.8571 ^{+42.9%}	0.9525 ^{+44.1%*}	0.9527 ^{+12.0%*}
c10mop1_0	0.7143 ^{+11.1%}	0.8298 ^{+14.1%}	0.8926 ^{+1.7%}
c10mop3_0	0.1429 ^{-71.4%*}	0.6335 ^{+22.6%}	0.7922 ^{-1.2%}
c10mop5_0	0.8571 ^{+114.3%*}	0.7626 ^{+32.5%*}	0.8887 ^{+9.9%*}
c10mop8_0	0.2857 ^{-53.5%*}	0.6138 ^{-6.5%}	0.7992 ^{-7.9%*}
citysegmop3_0	0.5714 ^{+11.1%}	0.5188 ^{-16.3%}	0.8077 ^{-2.5%}
in1kmop1_0	0.5714 ^{+29.0%}	0.6514 ^{+11.0%}	0.8324 ^{+3.1%}
in1kmop4_0	0.5714 ^{+17.6%}	0.6989 ^{+15.7%}	0.8570 ^{+3.3%}
in1kmop7_0	1.0000 ^{+55.6%*}	0.7249 ^{+10.5%}	0.9134 ^{+6.5%}
Average	0.5357^{-1.7%}	0.6818^{+8.9%}	0.8455^{+0.9%}

TABLE II: NDCG performance of L2RAC compared to random ranking for next release problem. The formate is the same as Table I.

Dataset	NDCG@1	NDCG@3	NDCG
nrp-e1	1.0000 ^{+62.8%*}	0.8788 ^{+33.4%*}	0.9610 ^{+11.8%*}
nrp-e2	1.0000 ^{+55.6%*}	0.9247 ^{+38.9%*}	0.9763 ^{+13.3%*}
nrp-e3	1.0000 ^{+66.7%*}	0.9247 ^{+43.1%*}	0.9763 ^{+14.5%*}
nrp-e4	1.0000 ^{+42.9%*}	0.9247 ^{+35.9%*}	0.9763 ^{+12.6%*}
nrp-g1	1.0000 ^{+125.8%*}	0.9247 ^{+52.8%*}	0.9763 ^{+18.9%*}
nrp-g2	1.0000 ^{+59.1%*}	0.8788 ^{+39.0%*}	0.9610 ^{+12.6%*}
nrp-g3	1.0000 ^{+94.4%*}	0.8848 ^{+49.0%*}	0.9630 ^{+16.3%*}
nrp-g4	1.0000 ^{+105.9%*}	0.8048 ^{+21.1%*}	0.9363 ^{+11.3%*}
nrp-m1	1.0000 ^{+79.5%*}	0.9548 ^{+57.9%*}	0.9820 ^{+18.0%*}
nrp-m2	1.0000 ^{+75.0%*}	0.9247 ^{+47.7%*}	0.9763 ^{+16.4%*}
nrp-m3	1.0000 ^{+45.8%*}	0.8803 ^{+26.1%*}	0.9621 ^{+10.6%*}
nrp-m4	1.0000 ^{+105.9%*}	0.9247 ^{+56.0%*}	0.9763 ^{+18.7%*}
nrl1	1.0000 ^{+112.1%*}	0.8660 ^{+44.8%*}	0.9647 ^{+18.1%*}
nrl2	0.2857 ^{-54.5%*}	0.6260 ^{-1.3%}	0.8158 ^{-3.8%}
nrl3	1.0000 ^{+84.2%*}	0.8428 ^{+38.3%*}	0.9490 ^{+13.7%*}
nrl4	1.0000 ^{+62.8%*}	0.9247 ^{+53.7%*}	0.9763 ^{+16.7%*}
nrl5	0.4286 ^{-30.2%*}	0.6877 ^{+7.2%}	0.8494 ^{-0.3%}
Average	0.9244^{+60.3%*}	0.8693^{+37.4%†}	0.9517^{+12.8%†}

TABLE III: NDCG performance of L2RAC compared to random ranking for software configuration tuning. The formate is the same as Table I.

Dataset	NDCG@1	NDCG@3	NDCG
dnn_adia	0.6667 ^{-2.4%}	0.7311 ^{+8.5%}	0.8930 ^{+2.7%}
dnn_adia_reverse	1.0000 ^{+59.1%*}	0.7545 ^{+18.0%}	0.9456 ^{+10.9%*}
dnn_coffee	0.1429 ^{-70.6%*}	0.6335 ^{+7.1%}	0.7922 ^{-3.1%}
dnn_coffee_reverse	0.5714 ^{-7.0%}	0.5762 ^{-18.5%*}	0.8458 ^{-2.3%}
dnn_dsr	0.5000 ^{-3.2%}	0.7193 ^{+17.8%}	0.8646 ^{+3.8%}
dnn_dsr_reverse	0.7143 ^{+13.6%}	0.7464 ^{+4.5%}	0.9118 ^{+4.7%}
dnn_sa	0.5000 ^{-18.9%}	0.4269 ^{-37.8%*}	0.7798 ^{-9.1%*}
dnn_sa_reverse	1.0000 ^{+180.0%*}	0.9901 ^{+97.2%*}	0.9906 ^{+26.2%*}
llvm	0.2857 ^{-48.7%*}	0.6711 ^{+13.2%}	0.8093 ^{-1.9%}
llvm_reverse	0.1429 ^{-70.6%*}	0.4355 ^{-28.5%*}	0.7524 ^{-9.2%*}
lrzip	1.0000 ^{+75.0%*}	0.8020 ^{+28.6%*}	0.9553 ^{+13.6%*}
lrzip_reverse	0.8571 ^{+46.3%*}	0.9050 ^{+42.6%*}	0.9396 ^{+11.9%*}
mariadb	0.8571 ^{+62.2%*}	0.6120 ^{-0.6%}	0.8570 ^{+4.2%}
mariadb_reverse	1.0000 ^{+62.8%*}	0.7626 ^{+13.1%}	0.9107 ^{+6.7%*}
mongodb	0.8571 ^{+33.3%*}	0.8217 ^{+19.1%}	0.9330 ^{+7.7%*}
mongodb_reverse	0.8571 ^{+42.9%*}	0.8100 ^{+12.7%}	0.9147 ^{+5.1%}
storm_rs	0.8571 ^{+100.0%*}	0.9722 ^{+60.8%*}	0.9611 ^{+16.2%*}
storm_rs_reverse	0.4286 ^{-23.1%}	0.5959 ^{+2.9%}	0.8194 ^{-1.3%}
storm_wc	0.5714 ^{+37.9%}	0.8495 ^{+61.5%*}	0.8953 ^{+12.0%*}
storm_wc_reverse	0.8571 ^{+53.8%*}	0.7823 ^{+16.6%}	0.9098 ^{+6.6%}
trimesh	0.7143 ^{-15.3%}	0.6039 ^{-15.2%*}	0.8641 ^{-3.7%}
trimesh_reverse	1.0000 ^{+105.9%*}	0.7070 ^{+19.9%}	0.9052 ^{+10.2%*}
vp9	0.8571 ^{+36.4%}	0.7545 ^{+7.5%}	0.9071 ^{+5.2%}
vp9_reverse	0.5714 ^{-2.4%}	0.5842 ^{-12.2%*}	0.8250 ^{-3.1%*}
x264	0.7143 ^{+19.0%}	0.9346 ^{+43.6%*}	0.9357 ^{+10.2%*}
x264_reverse	0.5000 ^{-6.2%}	0.5966 ^{-13.7%}	0.8315 ^{-2.7%}
Average	0.6932^{+22.2%}	0.7222^{+12.6%*}	0.8827^{+4.6%*}

TABLE IV: NDCG performance of L2RAC compared to random ranking for software effort estimation. The formate is the same as Table I.

Dataset	NDCG@1	NDCG@3	NDCG
china-train_fold1	0.4286 ^{-25.0%}	0.7742 ^{+16.2%*}	0.8622 ^{+1.4%}
china-train_fold2	0.8571 ^{+25.0%}	0.5842 ^{-19.9%*}	0.8663 ^{-1.7%}
china-train_fold3	0.5714 ^{+14.3%}	0.6317 ^{+6.2%}	0.8569 ^{+3.1%}
desharnais-train_fold1	1.0000 ^{+34.6%*}	0.9050 ^{+36.0%*}	0.9754 ^{+12.0%*}
desharnais-train_fold2	1.0000 ^{+94.4%*}	0.7921 ^{+29.3%*}	0.9473 ^{+13.9%*}
desharnais-train_fold3	1.0000 ^{+112.1%*}	0.6873 ^{+20.6%*}	0.9030 ^{+11.3%*}
finnish-train_fold1	0.2857 ^{-42.9%}	0.7088 ^{+16.1%}	0.8348 ^{+0.8%}
finnish-train_fold2	0.8571 ^{+140.0%*}	0.8575 ^{+43.3%*}	0.9180 ^{+14.3%*}
finnish-train_fold3	0.8571 ^{+53.8%*}	0.9050 ^{+40.2%*}	0.9413 ^{+10.4%*}
maxwell-train_fold1	0.8571 ^{+93.5%*}	0.9722 ^{+71.8%*}	0.9690 ^{+20.3%*}
maxwell-train_fold2	0.7143 ^{+13.6%}	0.6989 ^{-0.9%}	0.9021 ^{+4.3%}
maxwell-train_fold3	0.8571 ^{+76.5%*}	0.9525 ^{+56.3%*}	0.9502 ^{+15.1%*}
miyazaki-train_fold1	0.8571 ^{+36.4%}	0.6971 ^{+5.9%}	0.8819 ^{+2.6%}
miyazaki-train_fold2	0.4286 ^{-3.2%}	0.6792 ^{+11.8%}	0.8461 ^{+3.2%}
miyazaki-train_fold3	0.8571 ^{+46.3%*}	0.7823 ^{+20.9%*}	0.9295 ^{+9.2%*}
Average	0.7619 ^{+40.8%*}	0.7752 ^{+22.6%*}	0.9056 ^{+7.9%†}

TABLE V: NDCG performance of L2RAC compared to random ranking for software defect prediction. The formate is the same as Table I.

Dataset	NDCG@1	NDCG@3	NDCG
ant-1.7_J48	1.0000 ^{+45.8%*}	0.9247 ^{+31.9%*}	0.9855 ^{+12.3%*}
ant-1.7_KNN	1.0000 ^{+59.1%*}	0.8199 ^{+19.1%}	0.9480 ^{+9.8%*}
ant-1.7_LR	0.5714 ^{-4.8%}	0.8495 ^{+23.0%}	0.8963 ^{+4.2%}
ant-1.7_NB	1.0000 ^{+66.7%*}	1.0000 ^{+51.2%*}	0.9894 ^{+16.3%*}
camel-1.6_J48	1.0000 ^{+75.0%*}	1.0000 ^{+43.8%*}	0.9951 ^{+15.8%*}
camel-1.6_KNN	0.1429 ^{-71.4%*}	0.6335 ^{+11.2%}	0.7941 ^{-2.8%}
camel-1.6_LR	1.0000 ^{+94.4%*}	0.9901 ^{+69.8%*}	0.9851 ^{+19.7%*}
camel-1.6_NB	1.0000 ^{+40.0%*}	0.9901 ^{+46.9%*}	0.9832 ^{+12.1%*}
ivy-2.0_J48	0.5714 ^{+17.6%}	0.6989 ^{+11.8%}	0.8812 ^{+5.6%*}
ivy-2.0_KNN	0.8571 ^{+50.0%*}	0.9722 ^{+46.0%*}	0.9747 ^{+15.0%*}
ivy-2.0_LR	1.0000 ^{+75.0%*}	1.0000 ^{+56.4%*}	0.9894 ^{+16.7%*}
ivy-2.0_NB	1.0000 ^{+84.2%*}	1.0000 ^{+55.0%*}	0.9968 ^{+18.1%*}
jedit-4.0_J48	0.1429 ^{-77.3%*}	0.4059 ^{-36.2%*}	0.7323 ^{-13.5%*}
jedit-4.0_KNN	1.0000 ^{+25.0%}	0.7348 ^{+1.8%}	0.9197 ^{+2.8%}
jedit-4.0_LR	1.0000 ^{+62.8%*}	0.8871 ^{+30.6%*}	0.9797 ^{+13.5%*}
jedit-4.0_NB	1.0000 ^{+84.2%*}	1.0000 ^{+62.8%*}	0.9899 ^{+18.9%*}
lucene-2.4_J48	0.1429 ^{-78.3%*}	0.6335 ^{-12.0%*}	0.7946 ^{-9.4%*}
lucene-2.4_KNN	1.0000 ^{+55.6%*}	0.8952 ^{+40.6%*}	0.9659 ^{+13.1%*}
lucene-2.4_LR	1.0000 ^{+48.9%*}	0.9427 ^{+45.5%*}	0.9777 ^{+14.1%*}
lucene-2.4_NB	1.0000 ^{+94.4%*}	1.0000 ^{+62.8%*}	0.9975 ^{+20.0%*}
poi-3.0_J48	0.4286 ^{+0.0%}	0.6416 ^{-8.2%}	0.8227 ^{-2.7%}
poi-3.0_KNN	1.0000 ^{+32.1%*}	1.0000 ^{+43.7%*}	0.9894 ^{+13.3%*}
poi-3.0_LR	0.8571 ^{+62.2%*}	0.9722 ^{+62.0%*}	0.9647 ^{+16.2%*}
poi-3.0_NB	1.0000 ^{+204.3%*}	0.8952 ^{+58.1%*}	0.9634 ^{+20.3%*}
synapse-1.2_J48	0.4286 ^{-26.8%}	0.4238 ^{-39.2%*}	0.7681 ^{-11.0%*}
synapse-1.2_KNN	1.0000 ^{+125.8%*}	0.8298 ^{+42.9%*}	0.9610 ^{+17.8%*}
synapse-1.2_LR	1.0000 ^{+55.6%*}	0.8495 ^{+24.9%*}	0.9645 ^{+11.3%*}
synapse-1.2_NB	1.0000 ^{+75.0%*}	1.0000 ^{+56.5%*}	0.9961 ^{+17.1%*}
velocity-1.6_J48	0.8571 ^{+36.4%}	0.7446 ^{+15.9%}	0.8956 ^{+5.3%}
velocity-1.6_KNN	1.0000 ^{+84.2%*}	0.9901 ^{+46.3%*}	0.9906 ^{+15.5%*}
velocity-1.6_LR	1.0000 ^{+84.2%*}	1.0000 ^{+58.2%*}	0.9968 ^{+18.9%*}
velocity-1.6_NB	0.4286 ^{-26.8%}	0.7742 ^{+29.3%*}	0.8503 ^{+2.5%}
xalan-2.6_J48	1.0000 ^{+55.6%*}	1.0000 ^{+49.7%*}	0.9956 ^{+15.4%*}
xalan-2.6_KNN	0.8571 ^{+122.2%*}	0.8970 ^{+67.2%*}	0.9648 ^{+21.2%*}
xalan-2.6_LR	1.0000 ^{+62.8%*}	1.0000 ^{+43.4%*}	0.9968 ^{+15.1%*}
xalan-2.6_NB	0.4286 ^{-36.2%*}	0.7840 ^{+6.1%}	0.8776 ^{-1.3%}
xerces-1.4_J48	0.1429 ^{-75.0%*}	0.4534 ^{-23.8%}	0.7600 ^{-9.3%*}
xerces-1.4_KNN	1.0000 ^{+94.4%*}	0.9901 ^{+73.4%*}	0.9892 ^{+20.5%*}
xerces-1.4_LR	1.0000 ^{+118.8%*}	0.9427 ^{+57.0%*}	0.9750 ^{+18.7%*}
xerces-1.4_NB	0.5714 ^{-2.4%}	0.8593 ^{+39.9%*}	0.8982 ^{+6.6%*}
Average	0.8107 ^{+40.5%†}	0.8606 ^{+33.2%†}	0.9349 ^{+10.3%†}

TABLE VI: NDCG performance of L2RAC compared to random ranking for software product lines testing. The formate is the same as Table I.

Dataset	NDCG@1	NDCG@3	NDCG
7z	1.0000 ^{+94.4% *}	0.6873 ^{+8.5%}	0.9024 ^{+8.1% *}
Amazon	1.0000 ^{+125.8% *}	0.8020 ^{+46.0% *}	0.9445 ^{+16.7% *}
BerkeleyDBC	0.8571 ^{+36.4% *}	0.8833 ^{+32.4% *}	0.9441 ^{+10.1% *}
CocheEcologico	0.9571 ^{+45.7% *}	0.7499 ^{+4.3%}	0.9242 ^{+5.7%}
CounterStrikeSimpleFeatureModel	0.7143 ^{+61.3%}	0.7372 ^{+25.6% *}	0.8927 ^{+9.9% *}
DSSample	1.0000 ^{+52.2% *}	0.7921 ^{+12.8%}	0.9393 ^{+7.4% *}
Drupal	0.7143 ^{+28.2%}	0.8243 ^{+22.7% *}	0.9198 ^{+8.0% *}
Dune	0.2857 ^{-58.3% *}	0.5830 ^{-12.9%}	0.8034 ^{-8.4% *}
ElectronicDrum	0.7143 ^{+42.9% *}	0.7464 ^{+19.8% *}	0.9044 ^{+8.3% *}
HiPAcc	0.8571 ^{+93.5% *}	0.6852 ^{+19.8% *}	0.8915 ^{+10.1% *}
JHipster	1.0000 ^{+59.1% *}	0.9394 ^{+35.1% *}	0.9704 ^{+12.3% *}
JavaGC	0.5714 ^{-16.7%}	0.7434 ^{+14.1%}	0.8819 ^{+3.4%}
ModelTransformation	0.8571 ^{+53.8% *}	0.9572 ^{+53.1% *}	0.9688 ^{+15.7% *}
Polly	0.8286 ^{+48.7% *}	0.8108 ^{+24.7% *}	0.9295 ^{+9.6% *}
SPLSSimuelESPnP	0.5714 ^{-4.8%}	0.7530 ^{+29.4% *}	0.8785 ^{+5.1% *}
SmartHomev2.2	0.7143 ^{+78.6% *}	0.8811 ^{+51.2% *}	0.9300 ^{+14.6% *}
VP9	0.8286 ^{+93.3% *}	0.8514 ^{+49.1% *}	0.9373 ^{+16.3% *}
VideoPlayer	0.5714 ^{-2.4%}	0.7351 ^{+7.6%}	0.8803 ^{+2.0%}
WebPortal	0.4286 ^{-38.8% *}	0.6922 ^{-2.5%}	0.8500 ^{-3.0%}
lrzip	0.5714 ^{-11.1%}	0.5318 ^{-17.8%}	0.8055 ^{-5.3%}
x264	0.7000 ^{+28.9%}	0.7630 ^{+21.4% *}	0.8846 ^{+5.9% *}
Average	0.7497^{+32.8% *}	0.7690^{+20.3% †}	0.9040^{+7.1% †}

TABLE VII: NDCG performance of L2RAC compared to random ranking for the software project scheduling. The formate is the same as Table I.

Dataset	NDCG@1	NDCG@3	NDCG
10-10-skill-4-5	0.8571 ^{+27.7%}	0.9722 ^{+40.4% *}	0.9771 ^{+11.5% *}
10-10-skill-6-7	0.3571 ^{-37.5%}	0.6427 ^{+1.4%}	0.8251 ^{-2.3%}
10-15-skill-4-5	0.8571 ^{+66.7% *}	0.9722 ^{+61.5% *}	0.9720 ^{+16.8% *}
10-15-skill-6-7	0.7143 ^{-5.7%}	0.8570 ^{+16.8% *}	0.9298 ^{+4.7%}
10-5-skill-4-5	0.4857 ^{+9.7%}	0.6597 ^{+25.4%}	0.8619 ^{+7.2%}
10-5-skill-6-7	0.8571 ^{+30.4%}	0.6120 ^{-10.9%}	0.8646 ^{-0.2%}
20-10-skill-4-5	0.7143 ^{+66.7% *}	0.9257 ^{+67.1% *}	0.9293 ^{+15.2% *}
20-10-skill-6-7	0.9286 ^{+75.7% *}	0.9861 ^{+51.6% *}	0.9752 ^{+15.8% *}
20-15-skill-4-5	1.0000 ^{+79.5% *}	0.9901 ^{+69.8% *}	0.9863 ^{+19.4% *}
20-15-skill-6-7	0.7143 ^{+16.3%}	0.8396 ^{+25.4% *}	0.9103 ^{+6.8%}
20-5-skill-4-5	0.7143 ^{+6.4%}	0.9247 ^{+31.4% *}	0.9360 ^{+7.6% *}
20-5-skill-6-7	0.6857 ^{+37.1%}	0.8262 ^{+32.2% *}	0.8862 ^{+5.7%}
30-10-skill-4-5	1.0000 ^{+125.8% *}	0.9990 ^{+65.1% *}	0.9857 ^{+20.1% *}
30-10-skill-6-7	0.7143 ^{+11.1%}	0.8796 ^{+37.7% *}	0.9225 ^{+8.5% *}
30-15-skill-4-5	0.8571 ^{+46.3% *}	0.9525 ^{+40.4% *}	0.9485 ^{+10.7% *}
30-15-skill-6-7	0.7143 ^{+6.4%}	0.9247 ^{+48.2% *}	0.9247 ^{+8.9% *}
30-5-skill-4-5	0.8571 ^{+46.3% *}	0.9346 ^{+55.3% *}	0.9616 ^{+15.1% *}
30-5-skill-6-7	0.8143 ^{+23.9% *}	0.9164 ^{+31.7% *}	0.9505 ^{+9.3% *}
Average	0.7690^{+31.8% *}	0.8786^{+37.5% †}	0.9304^{+10.0% †}

TABLE VIII: NDCG performance of L2RAC compared to random ranking for third party library migration. The format is the same as Table I.

Dataset	NDCG@1	NDCG@3	NDCG
migrationRule10_ALL	0.2857 ^{-51.2%*}	0.4720 ^{-27.0%*}	0.7640 ^{-10.4%*}
migrationRule10_CO	0.4286 ^{+15.4%*}	0.4337 ^{-18.4%*}	0.7655 ^{-3.7%*}
migrationRule10_DS	0.4857 ^{-10.5%*}	0.6249 ^{+3.4%*}	0.8341 ^{-0.3%*}
migrationRule10_MS	0.4286 ^{-30.2%*}	0.6690 ^{-1.6%*}	0.8338 ^{-2.6%*}
migrationRule18_ALL	0.2857 ^{-50.0%*}	0.7019 ^{+12.3%*}	0.8233 ^{-2.9%*}
migrationRule18_CO	0.5429 ^{+0.0%*}	0.6337 ^{+22.2%*}	0.8254 ^{+2.2%*}
migrationRule18_DS	1.0000 ^{+72.4%*}	0.7227 ^{+14.7%*}	0.9183 ^{+10.5%*}
migrationRule18_MS	1.0000 ^{+112.1%*}	0.9931 ^{+56.8%*}	0.9809 ^{+18.7%*}
migrationRule1_ALL	1.0000 ^{+100.0%*}	0.9685 ^{+69.4%*}	0.9868 ^{+20.9%*}
migrationRule1_CO	0.7143 ^{+38.9%*}	0.9316 ^{+53.6%*}	0.9462 ^{+14.3%*}
migrationRule1_DS	0.4286 ^{-25.0%*}	0.7332 ^{+13.7%*}	0.8596 ^{+1.7%*}
migrationRule1_MS	0.6571 ^{+21.1%*}	0.7889 ^{+30.8%*}	0.9111 ^{+9.6%*}
migrationRule2_ALL	0.3429 ^{-27.3%*}	0.7100 ^{+17.8%*}	0.8414 ^{+2.8%*}
migrationRule2_CO	0.8571 ^{+71.4%*}	0.9663 ^{+55.5%*}	0.9713 ^{+16.5%*}
migrationRule2_DS	0.8571 ^{+53.8%*}	0.9647 ^{+58.5%*}	0.9763 ^{+16.9%*}
migrationRule2_MS	0.8714 ^{+41.9%*}	0.9020 ^{+39.4%*}	0.9602 ^{+13.0%*}
migrationRule3_ALL	0.5714 ^{+48.1%*}	0.8020 ^{+56.2%*}	0.8906 ^{+13.2%*}
migrationRule3_CO	1.0000 ^{+94.4%*}	0.8575 ^{+39.6%*}	0.9576 ^{+15.0%*}
migrationRule3_MS	0.8571 ^{+15.4%*}	0.8147 ^{+12.2%*}	0.9257 ^{+4.7%*}
migrationRule4_ALL	1.0000 ^{+89.2%*}	0.9427 ^{+54.6%*}	0.9784 ^{+17.1%*}
migrationRule4_CO	0.5571 ^{+69.6%*}	0.6020 ^{+5.9%*}	0.8282 ^{+3.8%*}
migrationRule4_DS	0.5714 ^{-31.0%*}	0.8573 ^{+20.3%*}	0.8973 ^{+0.4%*}
migrationRule4_MS	1.0000 ^{+62.8%*}	0.8952 ^{+36.7%*}	0.9462 ^{+9.9%*}
migrationRule5_ALL	0.5857 ^{+32.3%*}	0.5820 ^{-5.2%*}	0.8223 ^{-0.1%*}
migrationRule5_CO	0.2857 ^{-44.4%*}	0.5459 ^{-1.4%*}	0.7913 ^{-2.7%*}
migrationRule5_DS	0.9571 ^{+109.4%*}	0.9471 ^{+56.3%*}	0.9749 ^{+18.3%*}
migrationRule5_MS	0.3143 ^{-43.6%*}	0.5516 ^{-19.2%*}	0.7978 ^{-7.5%*}
migrationRule7_ALL	0.7714 ^{+68.8%*}	0.9382 ^{+60.6%*}	0.9541 ^{+17.3%*}
migrationRule7_CO	0.5714 ^{+21.2%*}	0.7396 ^{+28.3%*}	0.8732 ^{+6.8%*}
migrationRule7_DS	1.0000 ^{+133.3%*}	0.9486 ^{+69.8%*}	0.9798 ^{+21.4%*}
migrationRule7_MS	0.8286 ^{+48.7%*}	0.9357 ^{+53.1%*}	0.9571 ^{+14.2%*}
migrationRule8_ALL	1.0000 ^{+133.3%*}	0.9172 ^{+60.6%*}	0.9714 ^{+19.4%*}
migrationRule8_CO	0.8571 ^{+62.2%*}	0.8354 ^{+39.2%*}	0.9339 ^{+12.9%*}
migrationRule8_DS	0.5429 ^{-30.9%*}	0.6390 ^{-13.2%*}	0.8462 ^{-4.9%*}
migrationRule8_MS	1.0000 ^{+48.9%*}	0.9548 ^{+28.6%*}	0.9832 ^{+11.2%*}
Average	0.6988 ^{+30.1%†}	0.7864 ^{+27.3%†}	0.9002 ^{+7.8%†}

TABLE IX: NDCG performance of L2RAC compared to random ranking for workflow scheduling. The format is the same as Table I.

Dataset	NDCG@1	NDCG@3	NDCG
CyberShake_100	0.2429 ^{-63.8%*}	0.3879 ^{-43.8%*}	0.7451 ^{-14.6%*}
CyberShake_1000	0.9143 ^{+68.4%*}	0.9332 ^{+47.8%*}	0.9581 ^{+14.3%*}
CyberShake_30	0.1429 ^{-78.3%*}	0.5939 ^{-16.7%*}	0.7769 ^{-11.4%*}
CyberShake_50	1.0000 ^{+84.2%*}	0.7371 ^{+15.2%*}	0.9287 ^{+10.7%*}
Epigenomics_100	0.6571 ^{+31.4%*}	0.5219 ^{+0.2%*}	0.8137 ^{+0.6%*}
Epigenomics_24	1.0000 ^{+79.5%*}	0.7819 ^{+12.9%*}	0.9376 ^{+9.8%*}
Epigenomics_46	1.0000 ^{+133.3%*}	0.8124 ^{+42.3%*}	0.9370 ^{+15.5%*}
Epigenomics_997	0.6143 ^{-12.2%*}	0.7538 ^{+16.9%*}	0.8942 ^{+4.7%*}
Inspir_100	0.2857 ^{-48.7%*}	0.4599 ^{-27.2%*}	0.7606 ^{-8.8%*}
Inspir_1000	0.7143 ^{+16.3%*}	0.7695 ^{+19.9%*}	0.8935 ^{+4.9%*}
Inspir_30	0.8571 ^{+42.9%*}	0.9007 ^{+33.9%*}	0.9599 ^{+11.9%*}
Inspir_50	1.0000 ^{+79.5%*}	0.7921 ^{+28.9%*}	0.9320 ^{+11.4%*}
Montage_100	0.7143 ^{+61.3%*}	0.6710 ^{+8.2%*}	0.8702 ^{+5.3%*}
Montage_1000	0.8571 ^{+93.5%*}	0.9082 ^{+60.1%*}	0.9579 ^{+18.0%*}
Montage_25	0.8571 ^{+57.9%*}	0.9346 ^{+57.0%*}	0.9691 ^{+16.2%*}
Montage_50	1.0000 ^{+125.8%*}	0.8796 ^{+49.1%*}	0.9646 ^{+18.3%*}
Sipht_100	0.7143 ^{+35.1%*}	0.6476 ^{+12.5%*}	0.8585 ^{+4.5%*}
Sipht_1000	0.8571 ^{+76.5%*}	0.7446 ^{+28.9%*}	0.8968 ^{+10.0%*}
Sipht_30	0.4286 ^{-33.3%*}	0.4327 ^{-35.3%*}	0.7670 ^{-11.0%*}
Sipht_60	1.0000 ^{+100.0%*}	0.8189 ^{+30.3%*}	0.9491 ^{+14.2%*}
Average	0.7429 ^{+35.6%*}	0.7241 ^{+16.0%*}	0.8885 ^{+6.1%*}

TABLE X: NDCG performance of L2RAC compared to random ranking for web service composition. The formate is the same as Table I.

Dataset	NDCG@1	NDCG@3	NDCG
workflow_1	0.1429 ^{-73.7%*}	0.5484 ^{-14.4%}	0.7877 ^{-6.6%}
workflow_10	0.8571 ^{+46.3%}	0.8298 ^{+24.3%}	0.9164 ^{+7.0%}
workflow_10_reverse	0.7143 ^{+22.0%}	0.9346 ^{+46.3%*}	0.9326 ^{+10.6%*}
workflow_1_reverse	0.5714 ^{+14.3%}	0.5762 ^{-16.6%}	0.8450 ^{-0.7%}
workflow_2	0.5714 ^{+11.1%}	0.8593 ^{+50.5%*}	0.8959 ^{+8.6%*}
workflow_2_reverse	0.2857 ^{-41.2%}	0.3862 ^{-43.6%*}	0.7379 ^{-12.7%*}
workflow_3	0.2857 ^{-61.5%*}	0.6237 ^{-9.0%}	0.8138 ^{-7.4%}
workflow_3_reverse	0.7143 ^{+78.6%*}	0.6416 ^{+18.8%}	0.8787 ^{+10.0%*}
workflow_4	0.7143 ^{+35.1%}	0.7840 ^{+27.1%*}	0.9136 ^{+9.8%*}
workflow_4_reverse	0.5714 ^{+29.0%}	0.7742 ^{+37.9%*}	0.9001 ^{+11.1%*}
workflow_5	0.7143 ^{+25.0%}	0.6219 ^{-3.4%}	0.8425 ^{-0.5%}
workflow_5_reverse	0.7143 ^{+4.2%}	0.6496 ^{-5.7%}	0.8641 ^{-1.2%}
workflow_6	1.0000 ^{+62.8%*}	0.8396 ^{+30.1%*}	0.9488 ^{+11.7%*}
workflow_6_reverse	1.0000 ^{+32.1%}	0.7348 ^{+1.2%}	0.9271 ^{+4.4%}
workflow_7	0.8571 ^{+62.2%*}	0.8396 ^{+41.9%*}	0.9296 ^{+12.2%*}
workflow_7_reverse	1.0000 ^{+133.3%*}	0.8477 ^{+58.1%*}	0.9460 ^{+18.4%*}
workflow_8	1.0000 ^{+89.2%*}	0.8100 ^{+29.3%*}	0.9165 ^{+9.6%*}
workflow_8_reverse	1.0000 ^{+89.2%*}	0.8199 ^{+32.0%*}	0.9379 ^{+11.6%*}
workflow_9	0.8571 ^{+87.5%*}	0.8020 ^{+28.4%*}	0.9434 ^{+14.5%*}
workflow_9_reverse	0.2857 ^{-39.4%}	0.5959 ^{-2.3%}	0.8176 ^{-1.2%}
Average	0.6929^{+27.1%*}	0.7259^{+15.1%*}	0.8848^{+5.3%*}